COMPETENCY STANDARD

FOR

Spinning, Winding, Beaming and Twisting Operation

(JUTE SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

This document is available at:

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-SWT-01-G	Apply occupational health and safety (OHS) practices at workplace	1. Identify OHS policies and procedures 2. Apply personal health and safety practices 3. Report hazards and risks 4. Respond to emergencies	10
SICIP- JUTE-SWT-03-G	Work in a team environment	1. Identify team goals and processes. 2. Communicate and cooperate with team members. 3. Work as a team member 4. Solve problems as a team member	10
		Total Hour	**20**

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP- JUTE-SWT-01-S	Work in the jute sector	1. Identify the jute sector and its main activities 2. Understand basic roles in jute production processes 3. Recognize common tools and equipment used 4. Understand simple quality requirements in jute work	10
SICIP- JUTE-SWT-02-S	Apply green practices in Jute sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**20**

Occupation Specific Competencies (180 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-JUTE-SWT-01-O	Demonstrate knowledge on Spinning, Winding, Beaming and Twisting operation	1. Understand the spinning machine components 2. Understand winding machine components 3. Comprehend beaming procedures and setup 4. Understand twisting components and equipment 5. Recognize factors influencing yarn quality in spinning, winding, beaming, and twisting 6. Understand safety practices in spinning, winding, beaming, and twisting operations	15
SICIP-JUTE-SWT-02-O	Prepare and Operate Spinning Machine	1. Interpret process specifications for spinning 2. Prepare the spinning machine 3. Operate the spinning machine 4. Perform doffing and pitching up operations 5. Clean and maintain the spinning machine and workplace.	45
SICIP-JUTE-SWT-03-O	Prepare and Operate Winding Machine	1. Set up the winding machine 2. Operate spool winding 3. Operate cop winding 4. Monitor and control winding quality 5. Clear yarn faults and join ends 6. Clean and maintain the winding machine and workplace.	40
SICIP-JUTE-SWT-04-O	Perform Beaming Operations	1. Prepare the creel for warping 2. Set beaming machine parameters 3. Operate the beaming machine	35

Code	Unit of Competency	Elements of Competency	Duration (hours)
		4. Transfer beam to the weaver's beam (beaming) 5. Identify and correct warp faults	
SICIP-JUTE-SWT -05-O	Operate Twisting Machine	1. Prepare the twisting machine for operation 2. Set machine parameters 3. Operate the doubling and twisting machine 4. Operate reeling machine 5. Clean and maintain the twisting machine and workplace.	25
SICIP-JUTE-SWT -06-O	Monitor and Maintain Yarn Quality	1. Conduct in-process yarn quality checks 2. Identify and classify yarn defects 3. Take corrective action on quality deviations 4. Complete quality documentation	20
		Total Hours	**180**

COMPETENCY STANDARD: SPINNING, WINDING, BEAMING AND TWISTING OPERATION

Generic Competencies:

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-SWT-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply Occupational Health and Safety (OH&S) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures; applying personal health and safety practices; reporting hazards and risks; and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are followed and practiced. 2.2 **<u>Personal Protective Equipment (PPE)</u>** is selected and used. 2.3 Personal hygiene is maintained.
3. Report hazards and risks	3.1 **<u>Hazards and risks</u>** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 **<u>Emergency response plans and procedures</u>** are implemented. 4.3 **<u>First aid procedure</u>** is applied during emergency situations.

Range of Variables

Variable	Range May include but not limited to:
1. OHS policies	1.1 International OHS requirements 1.2 Bangladesh standards for OHS 1.3 Building Code 1.4 Fire Safety Rules and Regulations 1.5 Industry Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gas Mask 2.3 Gloves 2.4 Safety shoes 2.5 Helmet 2.6 Face mask 2.7 Overalls 2.8 Goggles and safety glasses 2.9 Ear plugs 2.10 Sun block 2.11 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Tourniquet 5.5 Applying CPR (Cardiopulmonary Resuscitation) 5.6 Taking appropriate medicine

Curricular Evidence Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response.

	1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Performing first aid. 2.7 Performing basic firefighting accessories using fire extinguishers 2.8 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Communication with peers, sub-ordinates and seniors in workplace. 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Reported incidents arising from hazards and risks to authority. 1.4 Emergency response plans and procedures are implemented. 1.5 Applied basic first aid procedure.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning

	2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-SWT-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal **<u>forms of communication</u>** are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood and valued. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, specifications and workplace procedures. 3.3 Team member's support with other members are made to ensure team achieves goals, awareness and requirements. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified.

	4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of variables:

Variables	**Range** (may include but not limited to)
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective interpersonal skills to interact with team members 1.5 Effective formal and informal forms of communication 1.6 Value of diversity in team functioning. 1.7 Correct use of workplace terminology 1.8 Team's duties, responsibilities, authorities, objectives and task requirements 1.9 Support mechanism to other members of team to ensure achievements of goals. 1.10 Methods of identifying current and potential problems faced by a team 1.11 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Identifying relationships within team and with other workers 2.4 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.5 Using formal and informal forms of communication 2.6 Understanding and valuing views and opinions of other team members 2.7 Performing tasks in accordance with organizational and team requirements, specifications and workplace procedures. 2.8 Supporting other members of the team to ensure team

	achieves goals, awareness and requirements. 2.9 Identifying current and potential problems faced by the team 2.10 Identifying solutions to the problem 2.11 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Manuals/Guidelines 4.3 Pens 4.4 Papers 4.5 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 communicated and cooperated with team members. 1.3 worked as a team member 1.4 solved problems as a team member
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies:

Unit of Competency: **WORK IN THE JUTE SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-SWT-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for work in the jute sector. It specifically includes the tasks of identifying the jute sector and its main activities, understanding basic roles in jute production processes, recognizing common tools and equipment used and understanding simple quality requirements in jute work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the jute sector and its main activities	1.1 The jute sector is identified within the context of national and global industries. 1.2 **<u>Main activities</u>** are recognized. 1.3 **<u>Different stages of jute value chain</u>** are described in simple terms. 1.4 Key workplaces in the jute sector are identified. 1.5 Importance of the jute sector in the economy and environment is understood.
2. Understand basic roles in jute production processes	2.1 **<u>Basic job roles in jute production</u>** are identified and described. 2.2 Simple production workflows and task sequences are understood. 2.3 Teamwork and coordination in production processes are identified. 2.4 Workplace safety roles and responsibilities are understood and followed.
3. Recognize common tools and equipment used	3.1 **<u>Common tools used in jute processing</u>** are identified correctly. 3.2 **<u>Basic machines and equipment used in production</u>** are recognized. 3.3 Functions of tools and equipment are understood in simple terms. 3.4 Basic maintenance needs of tools and equipment are recognized.
4. Understand simple quality requirements in jute work	4.1 Basic quality standards in jute work are identified. 4.2 **<u>Simple defects in jute products</u>** are recognized. 4.3 Importance of maintaining quality during production is understood.

	4.4 Work output is checked against basic quality expectations. 4.5 Good workmanship practices are followed to ensure quality.

Range of Variables

Variable	**Range** May include but not limited to:
1. Main activities	1.1 Cultivation and harvesting of jute 1.2 Retting and fibre extraction 1.3 Baling and grading 1.4 Spinning and yarn production 1.5 Weaving and fabric production 1.6 Dyeing, printing, and finishing 1.7 Packaging and export
2. Different stages of jute value chain	2.1 Raw jute production (farm level) 2.2 Fibre processing (retting, stripping, drying) 2.3 Baling and grading 2.4 Spinning into yarn 2.5 Weaving/knitting into fabric 2.6 Finishing and value addition 2.7 Distribution and export
3. Basic job roles in jute production	3.1 Raw jute handler 3.2 Machine operator 3.3 Quality checker/inspector 3.4 Dyeing and finishing worker 3.5 Packing and store assistant 3.6 Supervisor or line in-charge
4. Common tools used in jute processing	4.1 Scissors and cutting knives 4.2 Measuring tape and rulers 4.3 Hand hooks and tongs 4.4 Weighing scale 4.5 Hand brushes
5. Basic machines and equipment used in production	5.1 Carding machine 5.2 Drawing machine 5.3 Spinning machine 5.4 Looms (handloom/power loom) 5.5 Dyeing vats or tanks 5.6 Baling press machine
6. Simple defects in jute products	6.1 Uneven thickness or slub in yarn 6.2 Broken fibres or yarn ends 6.3 Colour variation or patchiness 6.4 Holes or tears in fabric 6.5 Misweave or weaving faults

Curricular Content Guide

1. Underpinning Knowledge	1.1 Overview of the jute sector within national and global industry context. 1.2 Main activities and stages of the jute value chain from raw material to finished products. 1.3 Basic job roles, responsibilities, and workflows in jute production processes. 1.4 Importance of teamwork, coordination, and workplace safety in production settings. 1.5 Identification and functions of common tools, machines, and equipment used in jute work. 1.6 Basic maintenance awareness for tools and equipment used in production. 1.7 Understanding of basic quality standards, product defects, and quality expectations. 1.8 Principles of good workmanship and maintaining quality during jute production processes.
2. Underpinning Skills	2.1 Understanding basic job roles and responsibilities in jute production processes 2.2 Following simple production workflows, task sequences, and teamwork practices 2.3 Applying workplace safety roles and responsibilities in production activities 2.4 Understanding basic functions and maintenance needs of tools and equipment 2.5 Applying quality awareness during production to meet expected standards 2.6 Checking work output against basic quality requirements 2.7 Following good workmanship practices to ensure consistent quality output
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified the jute sector and its main activities 1.2 understood basic roles in jute production processes 1.3 recognized common tools and equipment used 1.4 understood simple quality requirements in jute work
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN JUTE SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-JUTE-SWT-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in Jute sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.6 Basic concepts of environmental sustainability and green practices are explained clearly. 1.7 **<u>Environmental impacts of jute production</u>** processes are identified. 1.8 **<u>Resource efficiency concepts</u>** are interpreted. 1.9 Waste reduction, reuse, and recycling principles are described in relation to jute operations. 1.10 **<u>Eco-friendly practices in the jute sector</u>** are recognized and related to workplace activities.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw materials are used efficiently according to workplace requirements. 2.2 **<u>Unnecessary consumption of resources</u>** is identified and avoided during operations. 2.3 Machines and equipment are operated in energy-efficient modes where applicable. 2.4 Material wastage is minimized through careful handling and accurate measurement. 2.5 **<u>Resource-saving practices</u>** are applied where possible. 2.6 Leakage, spillage, and wastage of materials are promptly controlled and reported. 2.7 **<u>Workplace practices</u>** are followed.
3. Implement waste management practices	3.1 **<u>Types of waste</u>** generated in jute operations are identified correctly. 3.2 Waste materials are segregated according to workplace and environmental requirements. 3.3 **<u>Reusable and recyclable materials</u>** are separated from non-recyclable waste. 3.4 Waste is stored safely in designated containers or

	areas. 3.5 Waste disposal is carried out following environmental rules and organizational procedures. 3.6 Hazardous or chemical waste is handled carefully according to safety guidelines. 3.7 Waste reduction practices are applied to minimize environmental impact during operations.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Environmental impacts of jute production	1.1 Water pollution from retting and washing 1.2 Chemical discharge from dyeing and finishing 1.3 Solid waste generation (jute dust, fibre waste) 1.4 Air pollution from boilers and machines 1.5 Energy consumption impacts
2. Resource efficiency concepts	2.1 Water conservation techniques 2.2 Energy-efficient machine operation 2.3 Optimized chemical usage 2.4 Material loss reduction 2.5 Process efficiency improvement
3. Eco-friendly practices in the jute sector	3.1 Use of natural dyes or low-impact dyes 3.2 Biodegradable materials use 3.3 Compliance with environmental regulations 3.4 Proper effluent treatment practices 3.5 Energy-saving production methods
4. Unnecessary consumption of resources	4.1 Overuse of water or chemicals 4.2 Idle machine running 4.3 Excess raw material usage 4.4 Poor process planning or duplication of work
5. Resource-saving practices	5.1 Reuse of rinse water where applicable 5.2 Recycling of jute waste and scraps 5.3 Reusing packaging materials 5.4 Recovery of usable materials from waste
6. Workplace practices	6.1 Following environmental policies and SOPs 6.2 Proper waste segregation and disposal 6.3 Use of eco-friendly materials 6.4 Compliance with environmental regulations 6.5 Maintaining clean and safe work environment
7. Types of waste	7.1 Jute fibre waste and dust 7.2 Yarn and fabric off-cuts 7.3 Chemical waste (dyes, salts, auxiliaries) 7.4 Wastewater from processing 7.5 Packaging waste (poly bags, cartons)

	7.6 Sludge from effluent treatment
8. Reusable and recyclable materials	8.1 Jute fibre scraps for reprocessing 8.2 Reusable packaging materials 8.3 Recyclable plastics and paper 8.4 Recoverable chemical containers (if applicable)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Concepts of environmental sustainability, green practices, and eco-friendly workplace behavior. 1.2 Environmental impacts of jute production and related industrial processes. 1.3 Principles of resource efficiency, including water, energy, and raw material conservation. 1.4 Waste reduction strategies such as reuse, recycling, and minimizing material wastage. 1.5 Efficient use of machines and equipment with focus on energy-saving practices. 1.6 Identification, segregation, and safe handling of different types of waste in jute operations. 1.7 Proper waste storage, disposal methods, and compliance with environmental regulations. 1.8 Hazardous waste handling and application of workplace environmental safety procedures.
2. Underpinning Skills	2.1 Applying resource efficiency principles in workplace operations 2.2 Applying waste reduction, reuse, and recycling practices in jute production activities 2.3 Promoting eco-friendly practices in daily workplace tasks 2.4 Using water, energy, and raw materials efficiently according to workplace requirements 2.5 Minimizing unnecessary consumption and wastage of resources during operations 2.6 Operating machines and equipment in energy-efficient modes where applicable 2.7 Handling materials carefully to reduce wastage and improve accuracy 2.8 Controlling leakage, spillage, and resource wastage and reporting issues promptly 2.9 Segregating different types of waste generated in jute operations 2.10 Handling hazardous waste according to safety guidelines and regulations
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety

	3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimized resources use in the workplace 1.3 implemented waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation Specific Competencies (180 Hrs.)

Unit of Competency: DEMONSTRATE KNOWLEDGE ON SPINNING, WINDING, BEAMING AND TWISTING OPERATION	Nominal Duration: 15 Hrs.	Unit Code: SICIP-JUTE-SWT-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to demonstrate knowledge on spinning, winding, beaming and twisting operation. It specifically includes the tasks of understanding the spinning machine components, understanding winding machine components, comprehending beaming procedures and setup, understanding twisting components and equipment, recognizing factors influencing yarn quality in spinning, winding, beaming, and twisting and understanding safety practices in spinning, winding, beaming, and twisting operations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in bold and underlined are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand the Spinning Machine Components	1.1 **<u>Key components</u>** of spinning machines are identified 1.2 The role of each component in the spinning process are explained. 1.3 **<u>Different types of spinning techniques</u>** are described and their applications in Jute production
2. Understand Winding Machine Components	2. 1 **<u>Key components</u>** of Winding machines are identified 2. 2 The role of each component in the Winding process are explained. 2. 3 **<u>Different types of Winding techniques</u>** are described and their applications in Jute production
3. Comprehend Beaming Procedures and Setup	3. 1 Beaming process and its importance are explained. 3. 2 **<u>Setup procedures</u>** for beaming machines are described 3. 3 Factors that influence beam quality is identified. 3. 4 **<u>Beaming components</u>** are identified.
4. Understand Twisting Components and Equipment	4. 1 Purpose of twisting in yarn processing are explained. 4. 2 S-twist are identified and their applications in Jute production. 4. 3 Settings and adjustments made on twisting machines are describe to achieve specific yarn characteristics.
5. Recognize Factors Influencing Yarn Quality in Spinning, Winding, Beaming,	5. 1 **<u>Key factors</u>** that affect yarn quality in spinning, winding, beaming, and twisting are identified. 5. 2 Machine settings and operator skills are described.

and Twisting	5. 3 **Defects** in each stage are explain impact the overall quality and performance of the textile product.
6. Understand Safety Practices in Spinning, Winding, Beaming, and Twisting Operations	6. 1 Describe the common hazards associated with spinning, winding, beaming, and twisting operations (e.g., moving parts, electrical risks). 6. 2 Explain safety measures and practices to minimize these hazards (e.g., machine guarding, proper training, PPE). 6. 3 Identify emergency procedures in case of equipment malfunction, fire, or other safety incidents.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Key components	1.1 Spindle 1.2 Flyer 1.3 Bobbin 1.4 Tensioning Device 1.5 Drafting System 1.6 Yarn Guide 1.7 Twist Insertion
2. Different types of spinning	2. 1 Apron draft 2. 2 Slip draft 2. 3 Roving 2. 4 Gill spinning
3. Key components of Winding machines	**3.1 Cop** 1. Spindle 2. Bobbin spindle 3. Foot – step 4. Yarn Tensioning Device 5. Yarn Guide 6. Brand spring 7. Flat Spring 8. Porcelin guide 9. Porcelin rod 10. Porcelin runner 11. Cop Cone 12. Cop spindle top 13. Spiral pinion 14. Traverse Mechanism 15. Helical wheel 16. Helical pinion 3.2 **Spool** 1. Scroll roller 2. Adaptor slib

	3. Spool center 4. Lifter bracket 5. Detector rod 6. Brest plat 7. Conical spring 8. Tension disc
4. Different types of Winding techniques	4. 1 Parallel Winding 4. 2 Spiral Winding (Conical Winding)
5. Setup procedures	5. 1 Beam loading 5. 2 Tension control 5. 3 Beam formation
6. Beaming components	6. 1 Creel 6. 2 Creel arrangement 6. 3 Creel spindle 6. 4 Tension disc 6. 5 Reed 6. 6 Starch Tray 6. 7 Starch roller 6. 8 Starch tank 6. 9 Squising roller 6. 10 Adjusting roller 6. 11 Steam roller 6. 12 Pressing roller 6. 13 Delivery roller 6. 14 PIV gear 6. 15 Dry cylinder with steam 6. 16 Beam Flange
7. Key factors	7. 1 Tension 7. 2 Temperature 7. 3 Humidity 7. 4 Strength 7. 5 Quality ratio 7. 6 CV Percent
8. Defects	8. 1 Yarn breakage 8. 2 Uneven winding

Curricular Content Guide

1. Underpinning Knowledge	1.1 Components of Spinning Machines 1.2 Types of Spinning Machines 1.3 Key components of a winding machine 1.4 Package Formation 1.5 Understanding the beaming process 1.6 Types of Beaming Machines 1.7 Knowledge of twisting machine components 1.8 Twisting Process Impact

	1.9 Knowledge of key yarn quality parameters 1.10 Understand factors affecting yarn quality 1.11 safety standards and regulations in textile manufacturing 1.12 The safety mechanisms of spinning, winding, beaming, and twisting machines
2. Underpinning Skills	2.1 Setting up the spinning machine 2.2 Maintenance and Troubleshooting 2.3 Adjusting and setting the winding machine 2.4 Monitoring winding efficiency and quality 2.5 Setting up and adjusting beaming machines 2.6 Identifying and correcting defective yarn delivery 2.7 Setting up twisting machines and adjusting settings 2.8 Monitoring yarn twist quality 2.9 Adjusting machine parameters 2.10 Operate machinery safely 2.11 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Understand the spinning machine components 1.2 Understand winding machine components 1.3 Comprehend beaming procedures and setup 1.4 Understand twisting components and equipment 1.5 Recognize factors influencing yarn quality in spinning, winding, beaming, and twisting 1.6 Understand safety practices in spinning, winding, beaming, and twisting operations.

2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	**Nominal**	**Unit Code:**

PREPARE AND OPERATE SPINNING MACHINE	Duration: 45 Hrs.	SICIP-JUTE-SWT-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to prepare and operate spinning machine. It specifically includes the tasks of interpreting process specifications for spinning, preparing the spinning machine, operating the spinning machine, performing doffing and pitching up operations and cleaning and maintain the spinning machine and workplace. .		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret process specifications for spinning	1.1 Process specification sheet is read 1.2 **<u>Process parameters</u>** are identified. 1.3 Can capacity/arrangement and bobbin requirements are confirmed against the production plan.
2. Prepare the spinning machine	2.1 **<u>Machine components</u>** are checked and adjusted 2.2 Can is loaded with the correct bobbins in the specified sequence. 2.3 Bobbin rail arrangement are checked and set to specification. 2.4 Machine is threaded up correctly. 2.5 Machine parameters (speed, draft, twist) are set according to the process specification.
3. Operate the spinning machine	3. 1 Machine is started following safe start-up procedure. 3. 2 Yarn is monitored continuously for breakages, count variation and twist irregularities. 3. 3 **<u>Yarn faults</u>** are detected 3. 4 Broken ends are pieced up promptly using correct piecing technique to minimize waste. 3. 5 **<u>Abnormal conditions</u>** are identified and corrected.
4. Perform doffing and pitching up operations	4. 1 Doffing is carried out at the correct package build using manual or automatic **<u>doffing procedure</u>**. 4. 2 Full bobbins are removed and placed on the empty bobbin without damage. 4. 3 Empty bobbins are placed on spindles correctly and the machine is restarted.
5. Clean and maintain the spinning machine and workplace.	5. 1 Flyer and fibre accumulation is cleaned from machine components at specified intervals. 5. 2 Travellers are replaced at the correct frequency as per the maintenance schedule. 5. 3 **<u>Cleaning activities</u>** and Maintenance activities are recorded in the machine log. 5. 4 Workplace is clean and maintained according to requirements.

Range of Variables

Variable	Range (May include but not limited to):
1. Process parameters	1.1 Yarn count-Lbs/Spindle (Tex) 1.2 Twist per inch/metre (TPI/TPM) 1.3 Twist multiplier (TM) 1.4 Spindle speed (RPM) 1.5 Draft ratio 1.6 Wharves' diameter
2. Machine components	4.1 Retaining rollers 4.2 Drafting rollers 4.3 Brest plate 4.4 Apron 4.5 Hook 4.6 Bobbin rail 4.7 Flyer 4.8 Conductor 4.9 Delivery roller 4.10 Spindle 4.11 Bobbin 4.12 Thread guide
3. Yarn faults	3.1 Thick places 3.2 Thin places 3.3 Slubs 3.4 End breaks 3.5 Twist variation 3.6 Count variation 3.7 Irregularity 3.8 CV percentage 3.9 Quality ratio 3.10 Strength 3.11 Hari
4. Abnormal conditions	4.1 Traveller flyers 4.2 Flyers eye breaks 4.3 Yarn detector 4.4 Builder motion 4.5 Bobbin traverse 4.6 Felt bob
5. Doffing procedure	5.1 Manual doffing 5.2 Semi-automatic doffing 5.3 Automatic doffing with doffing time
6. Cleaning activities	6.1 Cleaning of drafting zone

	6.2 Bobbin rail 6.3 Flyer replacement 6.4 Roller cleaning 6.5 Pully/Motor side 6.6 Gear side

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of ring spinning including drafting, twisting and winding 1.2 Types of yarn count systems and twist calculations 1.3 Machine components, their function and common faults 1.4 Correct piecing and doffing procedures 1.5 Traveller selection and replacement criteria 1.6 Quality monitoring during spinning 1.7 Cleaning and maintenance requirements
2. Underpinning Skills	2.1 Reading and applying process specification sheets 2.2 Loading creels and threading up ring frames 2.3 Setting machine parameters correctly 2.4 Piecing broken ends using correct technique 2.5 Monitoring yarn quality during operation 2.6 Performing manual and automated doffing 2.7 Replacing travellers and cleaning machine components
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of	Assessment required evidence that the candidate:

Competency	1.1 Interpret process specifications for spinning 1.2 Prepare the spinning machine 1.3 Operate the spinning machine 1.4 Perform doffing and pitching up operations 1.5 Clean and maintain the spinning machine and workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PREPARE AND OPERATE WINDING MACHINE**	**Nominal Duration:** 40 Hrs**.**	**Unit Code:** SICIP-JUTE-SWT-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to prepare and operate winding machine. It specifically includes the tasks of setting up the winding machine, operating spool winding, operating cop winding, monitoring and controlling winding quality, clearing yarn faults and join ends and cleaning and maintaining the winding machine and workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Set up the winding machine	1.1 **Winding parameters** are confirmed from the process specification. 1.2 Bobbin is loaded with the correct **input packages**. 1.3 Bobbin spindle and scroll roller are checked and set for the correct package size. 1.4 Yarn guides, tensioners and disc (if applicable) are positioned correctly.
2. Operate Spool winding machine	2.1. Spool winding machine is set up to meet production requirements. 2.2. The yarn is threaded through the winding system 2.3. The winding process is monitored and necessary adjustments are made to maintain consistent yarn placement and package size. 2.4. **Defects** are identified and corrected to minimize waste and downtime. 2.5. The finished spool is removed from the machine.
3. Operate Cop winding machine	3. 1 Cop winding machine is set up with correct **parameters** according to the job specifications. 3. 2 The yarn is threaded correctly through the winding system 3. 3 The cop winding process is monitored and irregularities are addressed immediately. 3. 4 The cop is wound to the required size and quality standards. 3. 5 Routine checks and maintenance are performed on the machine.
4. Monitor and control winding quality	4. 1 Package build, winding tension and traverse ratio are monitored continuously during operation. 4. 2 Yarn clearer cuts are observed and logged; splice quality is verified visually. 4. 3 **Defective cop and spool** are identified and removed.

5. Clear yarn faults and join ends	5. 1 Yarn breaks and clearer cuts are attended to promptly and yarn is joined using the pneumatic splicer. 5. 2 Knots are replaced with splices where required by quality specification. 5. 3 Faulty splices are removed and rejoined to achieve acceptable strength and appearance.
6. Clean and maintain the winding machine and workplace	3.1 **<u>Machine components</u>** are checked and adjusted. 3.2 Wastage and fibre accumulation is removed from tensioners, yarn guides and suction tubes at specified intervals. 3.3 Tensioner discs and blocks are replaced when worn or depleted. 3.4 Waste materials are collected and disposed of according to workplace standard operating procedures (SOP). 3.5 Tools and equipment are cleaned and restored. 3.6 Machines are cleaned, maintained. 3.7 Maintenance activities are logged in the machine maintenance record. 3.8 Workplace area is cleaned and organized following established workplace procedures.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Winding parameters	1.1 Winding speed (m/min) 1.2 Traverse ratio 1.3 Package density 1.4 Winding tension (cN) 1.5 Cone weight 1.6 Yarn clearer sensitivity settings
2. Defects	2. 1 Uneven tension 2. 2 Yarn breakage
3. Parameters	3. 1 Tension 3. 2 Spee 3. 3 Package size
4. Machine components	4. 1 Bobbin magazine / creel 4. 2 Tensioner 4. 3 Yarn clearer (electronic) 4. 4 Splicer 4. 5 Wax block 4. 6 Traverse drum 4. 7 Package holder

5. Defective cop and spool	5. 1 Ribboning 5. 2 Soft package (low density) 5. 3 Off-centre winding 5. 4 Broken splice 5. 5 Tight end Contamination
6. Input packages	6. 1 Spool winding 6. 2 Twisting bobbins 6. 3 Hanks (if rewinding) 6. 4 Cops winding

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of cone winding and package build 1.2 Winding parameters and their effects on package quality 1.3 Electronic yarn clearer operation and fault classification 1.4 Pneumatic splicer operation and splice quality requirements 1.5 Common winding defects, causes and corrective actions 1.6 Machine components and maintenance requirements
2. Underpinning Skills	2.1 Setting winding parameters from process specification 2.2 Loading input packages and threading the yarn path 2.3 Setting electronic yarn clearer sensitivity 2.4 Operating the pneumatic splicer correctly 2.5 Monitoring package build and winding quality 2.6 Identifying and removing defective packages 2.7 Cleaning and maintaining machine components
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Process specification sheet 4.3 Input bobbins/packages 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Set up the winding machine 1.2 Operated Spool winding 1.3 Operated Cop winding 1.4 Monitored and control winding quality 1.5 Cleared yarn faults and join ends 1.6 Cleaned and maintain the winding machine and workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **Perform Beaming Operations**	**Nominal Duration:** 35 Hrs.	**Unit Code:** SICIP-JUTE-SWT-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform beaming operations. It specifically includes the tasks of preparing the creel for warping, setting beaming machine parameters, operating the beaming machine and transferring beam to the weaver's beam (beaming).		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the creel for warping	1.1 Creel capacity and arrangement are planned according to the weaving design and process specification. 1.2 Spool are loaded onto the creel spindle in the correct position as per the colour/yarn sequence plan. 1.3 Yarn guides, tensioners and lease rod, reed and starch are checked and set to the correct position. 1.4 Yarn breaks on the creel are pieced up and tensioners are verified for uniform tension. 1.5 **<u>Warping methods</u>** is selected according to job requirements
2. Set Beaming machine parameters	1.1 **<u>Beaming machine parameters</u>** are set on the machine control panel. 1.2 Counter and length measuring device are set to zero and verified against production order. 1.3 Reed dent width and yarn arrangement through the reed are confirmed against the design specification
3. Operate the Beaming machine	1.1 **<u>Machine components</u>** are checked and adjusted. 1.2 Machine is started following correct start-up procedure. 1.3 Warp build is monitored continuously for yarn breaks, crossed ends and uneven tension. 1.4 Broken ends are pieced up promptly using correct tying method to maintain warp integrity. 1.5 Lease is inserted at specified intervals to maintain end separation.
4. Transfer beam to the weaver's beam	4.1 Warp sections or direct beam are transferred to the weaver's beam under controlled and uniform tension. 4.2 Beam width and warp length are verified against the production order on completion. 4.3 End leakage and beam flanges are checked and

	corrected before sending beam to loom. 4.4 Beam identification tag is completed with required production data.
5. Identify and correct warp faults	5. 1 **<u>Warp faults</u>** are identified. 5. 2 Corrective action is taken immediately and supervisor is informed if the fault cannot be corrected during production.

Range of Variables

Variable	**Range** (May include but not limited to):
1. Warping methods	1. 1 Direct (beam) warping 1. 2 Sectional warping
2. Warping parameters	2. 1 Warp length (metres) 2. 2 Number of ends 2. 3 Beam width (cm) 2. 4 Sectional width 2. 5 Warping speed (m/min) 2. 6 Winding tension
3. Machine components	3. 1 Creel 3. 2 Tensioner 3. 3 Yarn guide 3. 4 Lease reed 3. 5 Measuring roller 3. 6 Warping drum / beam barrel 3. 7 Flanged beam
4. Warp faults	4. 1 Missing ends 4. 2 Crossed ends 4. 3 Uneven tension 4. 4 Knots within warp 4. 5 Wrong yarn / colour sequence 4. 6 Damaged yarn

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of direct and sectional warping 1.2 Beam specifications and warp calculations 1.3 Creel loading patterns and tensioner setting 1.4 Warp fault identification and classification 1.5 Transfer of warp to weaver's beam procedures 1.6 Quality requirements for a warped beam 1.7 Safety requirements for warping operations
2. Underpinning Skills	2.1 Planning and loading the creel for warping 2.2 Setting warping machine parameters from specification

	2.3 Threading yarn through the reed and yarn guides 2.4 Monitoring warp build for breaks and faults 2.5 Piecing broken ends using correct tying method 2.6 Inserting the lease correctly 2.7 Transferring warp to weaver's beam under controlled tension 2.8 Completing beam identification documentation
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Different parts of sweater 4.3 Standard operating procedure 4.4 Linking machine 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared the creel for warping 1.2 set Beaming machine parameters 1.3 operated the Beaming machine 1.4 transferred beam to the weaver's beam (beaming). 1.5 identified and correct warp faults
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **Operate Twisting Machine**	**Nominal Duration:** 25 Hrs.	**Unit Code:** SICIP-JUTE-SWT-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate twisting machine. It specifically includes the tasks of preparing the twisting machine for operation, setting machine parameters, operating the twisting machine, operating reeling machine and cleaning and maintaining the twisting machine and workplace.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the twisting machine for operation	1.1 **Twisting machine** is selected as per job requirements. 1.2 Process specification is read and twist per inch/metre (TPI/TPM), twist direction (S or Z), yarn count and machine speed are confirmed. 1.3 Input packages (spool, bobbins) are selected and inspected for correct yarn count and absence of defects. 1.4 Packages are loaded onto the machine creel or spindle assembly in the correct position. 1.5 Yarn is threaded correctly through the yarn path including tensioners, guides and the spindle assembly.
2. Set machine parameters	2.1 **Process parameters** are set on the machine control panel or gear selection system. 2.2 **Input packages** are set and checked. 2.3 Twist direction is confirmed by checking the yarn path configuration on the machine. 2.4 Package take-up tension and traverse motion are set for the correctly.
3. Operate the twisting machine	3. 1 Machine is started following the correct safe start-up procedure. 3. 2 Yarn is monitored for end breaks, twist variation, irregularities and package defects during production. 3. 3 Broken ends are pieced up using correct splicing or knotting method as specified. 3. 4 **Output packages** is periodically verified using a twist tester against the specification. 3. 5 **Yarn faults during twisting** is identified and corrective measures are taken.
4. Operate reeling machine	4. 1 Reeling machine is set up correctly to meet the production requirements. 4. 2 The yarn is threaded properly through the reeling

	system. 4. 3 The reeling process is monitored, ensuring consistent and uniform reel winding. 4. 4 The reel is wound to the specified size and weight. 4. 5 Routine maintenance checks are conducted on the reeling machine.
5. Clean and maintain the twisting machine and workplace.	5. 1 Fibre accumulation is removed from spindle assemblies, tensioners and yarn guides at specified intervals. 5. 2 Spindle are checked and topped up as per the maintenance schedule. 5. 3 Maintenance records are updated after each maintenance activity. 5. 4 Workplace is cleaned and maintained as required.

Range of Variables

Variable	Range (May include but not limited to):
1. Twisting machine types	1.1 Two-for-one (TFO) twister 1.2 Ring twister 1.3 Doubling frame
2. Process parameters	2.1 Twist per metre (TPM) 2.2 Twist direction (S or Z) 2.3 Spindle speed (RPM) 2.4 Delivery speed 2.5 Package weight 2.6 Winding tension
3. Input packages	3.1 Yarn cones 3.2 Ring spinning bobbins 3.3 Open-end bobbins 3.4 Plied yarn packages
4. Output packages	4.1 Twisted yarn cones 4.2 Twisted yarn cheeses 4.3 Twisted yarn bobbins for weaving or knitting
5. Yarn faults during twisting	5.1 End breaks 5.2 Twist variation (over-twist / under-twist) 5.3 Balloon irregularity 5.4 Tight or soft packages splice failures

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of yarn doubling and twisting (ring twisting and TFO twisting) 1.2 Twist direction (S and Z) and twist multiplier calculations 1.3 Machine components and their functions 1.4 Yarn path configuration for TFO and ring twisters 1.5 Effect of twist on yarn strength and fabric characteristics 1.6 Quality monitoring during twisting operations

	1.7 Maintenance requirements for twisting machines
2. Underpinning Skills	2.1 Reading and applying process specifications for twisting 2.2 Loading input packages and threading the yarn path correctly 2.3 Setting spindle speed, delivery speed and TPM on machine control 2.4 Verifying twist direction and level using a twist tester 2.5 Monitoring yarn quality during operation 2.6 Piecing broken ends using correct technique 2.7 Performing doffing and package labelling 2.8 Carrying out routine cleaning and maintenance
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Tools, equipment and facilities appropriate to the process or activities. 4.3 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared the twisting machine for operation 1.2 set machine parameters 1.3 operated the twisting machine 1.4 operated reeling machine 1.5 cleaned and maintain the twisting machine and workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **Monitor and Maintain Yarn Quality**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-JUTE-SWT-06-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to monitor and maintain yarn quality. It specifically includes the tasks of conducting in-process yarn quality checks, identifying and classifying yarn defects, taking corrective action on quality deviations and completing quality documentation.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Conduct in-process yarn quality checks	1. 1 Yarn samples are collected from production at specified intervals as per the quality control plan. 1. 2 **Quality tests** are performed using **quality instruments**. 1. 3 Test results are compared against quality specifications and acceptable quality levels (AQL).
2. Identify and classify yarn defects	2. 1 **Yarn defects** are identified by visual examination or instrument testing. 2. 2 Defects are classified and recorded on the quality record form. 2. 3 Defective packages or beams are segregated and labelled for review
3. Take corrective action on quality deviations	3. 1 Root cause of quality deviation is identified (machine, material or process related). 3. 2 Corrective action is taken within the **operator's authority** 3. 3 Deviations outside the operator's authority are reported promptly to the supervisor.
4. Complete quality documentation	4. 1 Quality test results, defect records and corrective actions are documented accurately on the required forms. 4. 2 **Quality documentation**/records are submitted to the supervisor or quality department at the specified frequency. 4. 3 Traceability information (batch number, machine number, shift) is recorded on all quality documents.

Range of Variables

Variable	Range (May include but not limited to):
1. Quality tests	1. 1 Yarn count measurement (Lbs/spy,Ne/Nm/tex) 1. 2 Twist per inch/metre (TPI/TPM) measurement 1. 3 Tensile strength test

	1. 4 CV test 1. 5 Moisture test 1. 6 Quality ratio 1. 7 Regularity test / Evenness test (Uster tester) 1. 8 Visual examination
2. Quality instruments	2. 1 Warp reel tester 2. 2 Balance 2. 3 Twist tester 2. 4 Tensile strength tester 2. 5 Uster evenness tester 2. 6 Visual examination board 2. 7 Diameter gauge 2. 8 Moisture meter
3. Yarn defects	3. 1 Contamination (colour / fibre type) 3. 2 Splice failures 3. 3 Thick places 3. 4 Thin places 3. 5 Slubs 3. 6 End breaks 3. 7 Twist variation 3. 8 Count variation 3. 9 Irregularity 3. 10 CV percentage 3. 11 Quality ratio 3. 12 Strength 3. 13 Hari
4. Operator's authority	4. 1 Adjusting tension 4. 2 Replacing traveller 4. 3 Adjusting clearer settings
5. Quality documentation	5. 1 In-process quality check sheet 5. 2 Defect record form 5. 3 Corrective action report 5. 4 Machine log 5. 5 Shift production report

Curricular Content Guide

1. Underpinning Knowledge	1. 1 Yarn quality parameters and specifications 1. 2 Yarn defect types, causes and effects on downstream processes 1. 3 Quality testing methods and instruments 1. 4 Acceptable quality levels (AQL) and control charts 1. 5 Root cause analysis techniques 1. 6 Quality documentation requirements and traceability
2. Underpinning Skills	2. 1 Collecting yarn samples correctly

	2. 2 Conducting yarn count, twist, strength and evenness tests 2. 3 Identifying and classifying yarn defects 2. 4 Segregating defective packages 2. 5 Identifying root causes of quality deviations 2. 6 Taking corrective actions within operator authority 2. 7 Completing quality records accurately
3. Underpinning Attitudes	3. 1 Commitment to occupational health and safety 3. 2 Promptness in carrying out activities 3. 3 Sincere and honest to duties 3. 4 Environmental concerns 3. 5 Eagerness to learn 3. 6 Tidiness and timeliness 3. 7 Respect for rights of peers and seniors in the workplace 3. 8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4. 1 Workplace (simulated or actual) 4. 2 Tools, equipment and facilities appropriate to the process or activities. 4. 3 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1. 1 conducted in-process yarn quality checks 1. 2 identified and classify yarn defects 1. 3 take corrective action on quality deviations 1. 4 completed quality documentation.
2. Methods of Assessment	Competency should be assessed by: 2. 1 Written test 2. 2 Practical demonstration 2. 3 Oral questioning 2. 4 Portfolio (Optional)
3. Context of Assessment	3. 1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3. 2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Spinning, Winding, Beaming and Twisting Machine Operation**
Number of Trainees:	30

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Internet connectivity	01
4.	White board	01
5.	Jute spinning machine (minimum 80/100 spindles)	1
6.	Winding machine (minimum 24 drums, both side)	1
7.	Beaming machine	1
8.	Twisting machine (minimum 80 spindles)	1
9.	Warp reel and electronic balance for yarn count measurement	3
10.	Twist tester	2
11.	Tensile strength tester	1
12.	Digital weighing scale	2
13.	Moisture meter	2

14.	Vernier caliper	2
15.	Pneumatic splicer (portable)	5
16.	Electronic yarn clearer unit (demonstration model)	1
17.	Creel (minimum 100 peg capacity for warping practice)	1
18.	Weaver's beam (flanged)	3
19.	Bobbin / package trolley	4
20.	Machine cleaning equipment (suction guns, cleaning brushes, blower machine)	3 each

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Spinning, Winding, Beaming and Twisting Machine Operation (Jute sector)** the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 30 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 30 trainees.